

(3)

(1)

Solution system for homogenous differential equation:

① Firstly we have to take $y = vx$ and then take previous process.

$$** (x^{\sqrt{}} + y^{\sqrt{}}) dx - 2xy dy = 0 \quad \text{--- ①}$$

solⁿ: let, $y = vx$

then,

$$\frac{dy}{dx} = x \frac{dv}{dx} + v$$

$$y dx - x dy = \sqrt{x^{\sqrt{}} + y^{\sqrt{}}} dx$$

$$\text{①} \Rightarrow x^{\sqrt{}} + y^{\sqrt{}} - 2xy \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^{\sqrt{}} + y^{\sqrt{}}}{2xy}$$

$$v + x \frac{dv}{dx} = \frac{x^{\sqrt{}} + x^{\sqrt{}} v^{\sqrt{}}}{2x^{\sqrt{}} v}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^{\sqrt{}}}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - v^{\sqrt{}}}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = -\frac{v^{\sqrt{}} - 1}{2v}$$

$$\Rightarrow \frac{2v}{v^{\sqrt{}} - 1} dv = -\frac{dx}{x}$$

$$\Rightarrow \log(v^{\sqrt{}} - 1) + \log x = \log e$$

$$\Rightarrow \ln \frac{y^{\sqrt{}} - x^{\sqrt{}}}{x}, x = \ln e$$

$$\Rightarrow \frac{y^{\sqrt{}} - x^{\sqrt{}}}{x} = e$$

$$\therefore y^{\sqrt{}} - x^{\sqrt{}} = ex. \quad (\text{Ans})$$

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$$(2) \frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$$

Soln: Let, $y = vx$

$$\frac{dy}{dx} = x \frac{dv}{dx} + v$$

$$\Rightarrow x \frac{dv}{dx} + v = v + \tan v$$

$$\Rightarrow x \frac{dv}{dx} = \tan v$$

$$\Rightarrow \cot v dv = \frac{dx}{x}$$

Now integrate both side we get,

$$\log \sin v = \log x + \log e$$

$$\Rightarrow \log \sin v = \log x e$$

$$\Rightarrow \sin v = x e$$

$$\therefore \sin \frac{y}{x} = x e. \quad (\text{Ans})$$

$$(3) (x \sin \frac{y}{x} + y \cos \frac{y}{x}) dx + x \cos \frac{y}{x} dy = 0$$

Soln: Let, $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

The given eqn can be written as,

$$\frac{dy}{dx} = - \frac{x \sin \frac{y}{x} - y \cos \frac{y}{x}}{x \cos \frac{y}{x}}$$

$$\Rightarrow v + x \frac{dv}{dx} = - \frac{x \sin v + vx \cos v}{x \cos v}$$

$$\Rightarrow v + x \frac{dv}{dx} = - \frac{\sin v - v \cos v}{\cos v}$$

$$\Rightarrow x \frac{dv}{dx} = -v - \frac{\sin v - v \cos v}{\cos v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-v \cos v - \sin v + v \cos v}{\cos v}$$

$$\Rightarrow x \frac{dv}{dx} = - \frac{\sin v}{\cos v}$$

$$\Rightarrow \frac{\cos v}{\sin v} dv = - \frac{dx}{x}$$

$$\Rightarrow \log \sin v + \log x = \ln c$$

$$\Rightarrow x \sin v = \ln c \quad \therefore x \sin \left(\frac{y}{x} \right) = c. \quad (\text{Ans})$$

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$$4. ydx - xdy = \sqrt{x^2 + y^2} dx$$

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Equation Reducible to homogeneous:

$$(i) \frac{dy}{dx} = \frac{ax+by+c}{a_1x+b_1y+c_1} \quad \text{if } \frac{a}{a_1} \neq \frac{b}{b_1}$$

$$x = \alpha + h, \quad \alpha, \beta \text{ new variable}$$

$$y = \beta + k \quad h, k \text{ new constant.}$$

$$(x+y+1)dx - (2x+2y+1)dy = 0$$

$$(ii) \frac{a}{a_1} = \frac{b}{b_1} \quad \text{then, } \begin{aligned} a &= a_1 m \\ b &= b_1 m \end{aligned}$$

$$1. \frac{dy}{dx} = \frac{2x+y-2}{3x+y-3}$$

$$\text{Soln } \frac{dy}{dx} = \frac{2x+y-2}{3x+y-3} \quad \text{--- (i)}$$

$$\text{Here, } \frac{a}{a_1} \neq \frac{b}{b_1}$$

$$\text{Let, } x = \alpha + h, \quad y = \beta + k$$

$$dx = d\alpha, \quad dy = d\beta$$

$$\frac{d\beta}{d\alpha} = \frac{dy}{dx} \quad \text{--- (ii)}$$

$$\therefore \frac{d\beta}{d\alpha} = \frac{2(\alpha+h) + \beta + k - 2}{3(\alpha+h) + \beta + k - 3}$$

$$\frac{d\beta}{d\alpha} = \frac{2h + 2\alpha + \beta + k - 2}{3\alpha + 3h + \beta + k - 3}$$

$$\frac{d\beta}{d\alpha} = \frac{2\alpha + \beta + 2h + k - 2}{3\alpha + \beta + 3h + k - 3} \quad \text{--- (iii)}$$

we chose h and k such that,

$$2h + k - 2 = 0$$

$$3h + k - 3 = 0$$

$$h = 1, \quad k = 0$$

From, eqn (iii)

$$\frac{d\beta}{d\alpha} = \frac{2\alpha + \beta}{3\alpha + \beta} \quad \text{--- (iv)}$$

$$\text{Let, } \beta = v\alpha$$

$$\frac{d\beta}{d\alpha} = v + \alpha \frac{dv}{d\alpha} \quad \text{--- (v)}$$

From eqn (iv) and (v)

$$v + \alpha \frac{dv}{d\alpha} = \frac{2\alpha + v\alpha}{3\alpha + v\alpha}$$

$$\Rightarrow \alpha \frac{dv}{d\alpha} = \frac{2\alpha + v\alpha}{3\alpha + v\alpha} - v$$

$$\Rightarrow \alpha \frac{dv}{d\alpha} = \frac{2+v}{3+v} - v$$

(5)

$$\Rightarrow \frac{3+v}{v^{\sqrt{v+2v-2}}} dv = -\frac{d\alpha}{\alpha}$$

$$\Rightarrow \int \frac{3+v}{v^{\sqrt{v+2v-2}}} dv + \frac{d\alpha}{\alpha} = 0$$

$$\Rightarrow \int \frac{\left[\frac{1}{2}\left\{(2v+3)\right\}+2\right] dv}{v^{\sqrt{v+2v-2}}} + \frac{d\alpha}{\alpha} = 0$$

$$\Rightarrow \frac{1}{2} \int \frac{2v+2}{v^{\sqrt{v+2v-2}}} dv + 2 \int \frac{dv}{v^{\sqrt{v+2v-2}}} + \int \frac{d\alpha}{\alpha} = 0$$

$$\Rightarrow \frac{1}{2} \log(v^{\sqrt{v+2v-2}}) + 2 \int \frac{dv}{(v+1)^{\frac{2}{\sqrt{3}}}} + \int \frac{d\alpha}{\alpha} = 0$$

$$\Rightarrow \frac{1}{2} \log(v^{\sqrt{v+2v-2}}) + \frac{2}{\sqrt{3}} \log \frac{v+1-\sqrt{3}}{v+1+\sqrt{3}} + \log \alpha = \frac{1}{2} c$$

$$\Rightarrow \log(v^{\sqrt{v+2v-2}}) + \log \alpha^{\sqrt{v+2v-2}} + \frac{2}{\sqrt{3}} \log \frac{v+1-\sqrt{3}}{v+1+\sqrt{3}} = c$$

$$\therefore \log \left(\frac{v^{\sqrt{v+2v-2}}}{\alpha^{\sqrt{v+2v-2}}} + 2 \cdot \frac{v}{\alpha} - 2 \right) + \log \alpha^{\sqrt{v+2v-2}} + \frac{2}{\sqrt{3}} \log \frac{v+1-\sqrt{3}}{v+1+\sqrt{3}} = c$$

(Am)

$$(11) \frac{dy}{dx} = \frac{6x-2y-7}{3x-y+4} \quad \text{--- (1)}$$

Since here, $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

Let, $3x-y = v$

then, $3 - \frac{dy}{dx} = \frac{dv}{dx}$

$$\Rightarrow 3 - \frac{dv}{dx} = \frac{dy}{dx}$$

$$\therefore (1) \Rightarrow 3 - \frac{dv}{dx} = \frac{2v-7}{v+4}$$

$$\Rightarrow 3 - \frac{2v-7}{v+4} = \frac{dv}{dx}$$

$$\Rightarrow \frac{v+19}{v+4} = \frac{dv}{dx}$$

$$\Rightarrow dx = \frac{v+19}{v+4} dv$$

$$\Rightarrow dx = \left(1 - \frac{15}{v+19}\right) dv$$

$$\Rightarrow x = v - 15 \log(v+19) + c$$

$$\Rightarrow 3x - y - 15 \log(3x - y + 19) - x = c$$

$$\therefore 2x - y - 15 \log(3x - y + 19) = c$$

(Am)

⑥

$$\text{例 } (x+y+1)dx - (2x+2y+1)dy = 0$$